



[BIT703 Web Technologies](#) / [Contents](#) / Topic 5: Client-side web application development – HTML / Understanding HTML

HTML SVG images 30m

SVG stands for Scalable Vector Graphic, a type of image file. SVG elements are used to create vector-based graphics on the web that have an XML format.

SVGs are not pixel-based images – they are graphics that are curves, lines, rectangles and so on. For example, you could use an SVG to create a circle on your web page.

As they describe mathematical vectors, browsers are able to render SVG images to any size without pixelation. Browsers are also able to animate SVG features such as colour, size and position.

We can create and modify SVG images using any text editor; however, we can also employ vector-based editors like Adobe Illustrator and Sketch (a vector editor for Macs).

The following CodePens demonstrate a range of complex shapes and animations that can be created using SVGs. Take a moment to look at each (click on the hyperlinks to open the CodePens):

- [Border animation by Sean McCaffery \(hover to activate\)](#)
- [Heart animation by Nikolay Talanov](#)
- [Rainbow Rocket Man by Chris Gannon](#)

In this section, we will discuss how to create SVG images in HTML.

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How to create SVGs

To create an SVG, you must use the `svg` element, which acts as the container for the SVG (that is, you will specify the graphic to be built within this `svg` element). You will use the width and height attributes to specify the size of the graphic as you did for embedding images and videos.

Then you will specify the shape to be created. There are predefined shapes that can be used in web development, such as:

- `<circle>` for a circle
- `<rect>` for a rectangle
- `<polygon>` for a polygon.

This CodePen example (Example 5.21) shows how to create a circle using the `<svg>` element in HTML. Have a go at editing the code to see if you can change the colour of the circle and the border.

See if you can create a rectangle or a polygon using the `<svg>` element.



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HTML

Result

EDIT ON

LIVE

```
<html>
  <body>
    <svg height="150" width="150">
      <circle cx="20" cy="20" r="10"
stroke="black" stroke-width="1"
fill=#008080 />
    </svg>
  </body>
</html>
```

Resources1x0.5x0.25xRerun

Example 5.21 SVG circle



Using the `viewBox` attribute, you can define the position and dimensions of an SVG viewport – the visible part of an SVG image.

The `viewBox` attribute consists of a list of four numbers: min-x, min-y, width and height.

These coordinators define the rectangular area which is mapped to the bounds of the SVG

Skip to main content (that we can scale the SVG element).

The SVG format is another language to learn, and it is best to use a purpose-built tool to create the images if you are not planning on using one from an online provider like SVG Repo. Tools like Inkscape are freely available and great for creating custom SVG drawings. In many cases, you can draw diagrams using Draw.io and export them as SVG.

If we include a viewBox attribute to Example 5.21, we can define the position and dimension for the SVG viewport for the circle.

Example 5.22 shows how the circle viewport changes.



HTML

Result

EDIT ON

LIVE

```
<html>
  <body>
    <svg height="1500" width="1500"
viewBox="0 0 400 400">
      <circle cx="20" cy="20" r="10"
stroke="black" stroke-width="1"
fill=#008080 />
    </svg>
  </body>
</html>
```

Resources1x 0.5x 0.25xRerun

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Example 5.22 Larger circle



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☒ SVG icon

Non-Assessed

Status: Done

Reset ↺

1. Navigate to an SVG icon source (we have used [Alien Vector SVG Icon \(149\) - SVG Repo](#)) and copy the SVG code provided there. Embed the SVG code in the HTML.
2. Modify the SVG code directly in the HTML to change the colour using the **fill** attribute.
3. Set the height and width of the SVG directly in HTML. Ensure that the root `<svg>` element in the SVG code includes `height="1500"` and `width="1500"`. If not, add or update these attributes.
4. View the result in your browser.

☒ I have completed this activity

Your HTML + SVG results should look similar to the following, with your chosen icon and colour: Task: SVG icon.

Previous attempts ▼

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Dig deeper

To learn more about the SVG format and the different images you can create, explore these resources:

- SVG in HTML, which explains each element and attribute in the SVG code
- HTML SVG Graphics, which demonstrates how to create different shapes using the element
- SVG viewBox Attribute, which provides a detailed explanation of viewBox and how to use it along with several examples.



Thoughts on this page?

